

Mapping Slaughterhouse Supply Zones

Mapping Slaughterhouse Supply Zones: A Reproducible, Geographically Generic Walkthrough

A plain-language companion to Brandão Jr., Rausch, Munger & Gibbs (2023),
Land

, 12(9), 1782

A note on the data. Every figure, table, and number in this report is computed from a fictitious dataset built specifically for this repository. No confidential cattle-transit (GTA), rural property registry (CAR), or slaughterhouse record is used or reproduced anywhere in this document. The workflow reconstructs the

analytical logic

of Brandão Jr. et al. (2023,

Land

, 12(9), 1782) in open-source tools, not its empirical findings.

Executive summary

Cattle can travel through several properties before reaching a slaughterhouse, which makes it hard to know exactly where the animals a company buys actually came from. This report walks through a complete, open-source, geographically generic workflow that estimates the area, or **supply zone**, that feeds each slaughterhouse, and separates that area into farms that sell cattle directly to the plant from farms that sell only through an intermediary. The workflow can be pointed at any region in the world by naming it in the project configuration; real administrative boundaries are then fetched from OpenStreetMap or the public-domain Natural Earth dataset, both open and globally available, or a deterministic offline layout is used when neither can be reached. Using an illustrative synthetic dataset configured for Rondônia, Mato Grosso, Pará, across 3 regions, 12 slaughterhouses, and the years 2013-2018, the workflow finds that the average direct supply zone of a signatory plant (a plant that has joined a public sourcing commitment) covers about 1.9 million hectares, that roughly a quarter of that area is natural vegetation, and that extending monitoring to indirect suppliers or non-signatory plants would substantially widen coverage of the cattle trade at the cost of a much larger area to monitor. Every number below comes from fictitious data; the value of the exercise is the method, not the estimate.

1. Why supply zones matter

Companies, regulators, and civil-society groups that want to know whether cattle products are linked to deforestation face a basic geographic problem: purchase records name a slaughterhouse, not a location on a map. A supply zone translates the plant's likely catchment area for cattle into an explicit polygon, so it can be overlaid with land use, deforestation, and protected-area data to ask questions such as: how much of a plant's likely sourcing area still has forest cover, and how does that compare between plants that have signed a public sourcing commitment (referred to here as "signatory" plants, after

Brazil's Cattle Agreement, the case this workflow was originally built around) and plants that have not.

The original study built these zones from confidential cattle-transit permits (GTA) and rural property registrations (CAR) using proprietary ArcGIS routines. This repository asks a narrower question with synthetic stand-in data: can the same analytical logic be reproduced end to end with transparent, open-source tools, and does doing so change how the results should be read. Full methodological detail is in `docs/METHODS.md`; this report focuses on what each result means for a non-specialist reader.

2. Study design in brief

The workflow simulates 6 years of fictitious cattle-transit records, rural property boundaries, slaughterhouse locations, land use, and deforestation across 3 regions. It then:

1. Links each transit record to a property using deterministic matching rules.
2. Classifies slaughterhouses as eligible when they process more than 1,000 head per year and hold a sanitary inspection code, matching the thresholds used in the source study.
3. Identifies **direct suppliers** (properties that sell straight to an eligible plant) and **tier-1 indirect suppliers** (properties that sell to a direct supplier, using the same 16-head minimum transaction threshold).
4. Chooses, separately for every plant and year, the distance at which cattle-volume-weighted spatial clustering (Global Moran's I) peaks, and uses that distance to aggregate supplier properties into a single supply zone (an open-source analogue of the original ArcGIS Aggregate Polygons step).
5. Cross-tabulates the resulting zones against land use, deforestation, carbon density, and protected or military areas.

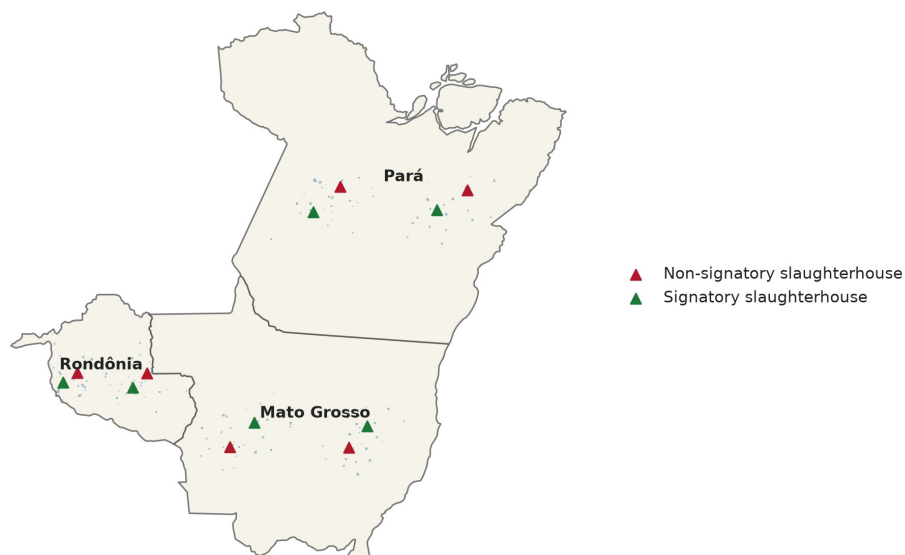


Figure 1. Synthetic study area, slaughterhouse locations, and linked properties.

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3. How large are supply zones, and how much do they overlap

Zone type	Zone-years observed	Mean annual area (million ha)	Median annual area (million ha)	Mean number of properties
Signatory direct	36	1.9	0.9	15.2
Signatory tier-1 indirect	36	0.1	0.0	2.9
Non-signatory direct	36	1.5	1.3	18.0
Non-signatory tier-1 indirect	36	0.1	0.0	3.6

Signatory plants have direct supply zones that are, on average, larger than their non-signatory counterparts in this synthetic dataset, largely because signatory plants tend to source from more properties. The zones are not mutually exclusive: a property can appear inside more than one plant's catchment, and different zone types partly cover the same territory.

Zone type A	Zone type B	Share of A inside B (%)	Share of B inside A (%)	Overlap index, Jaccard (%)
Signatory direct	Signatory tier-1 indirect	5.6%	65.2%	5.4%
Signatory direct	Non-signatory direct	54.3%	50.7%	35.6%
Signatory direct	Non-signatory tier-1 indirect	8.1%	58.6%	7.6%
Signatory tier-1 indirect	Non-signatory direct	70.1%	5.6%	5.5%
Signatory tier-1 indirect	Non-signatory tier-1 indirect	45.9%	28.5%	21.3%
Non-signatory direct	Non-signatory tier-1 indirect	11.7%	90.9%	11.6%

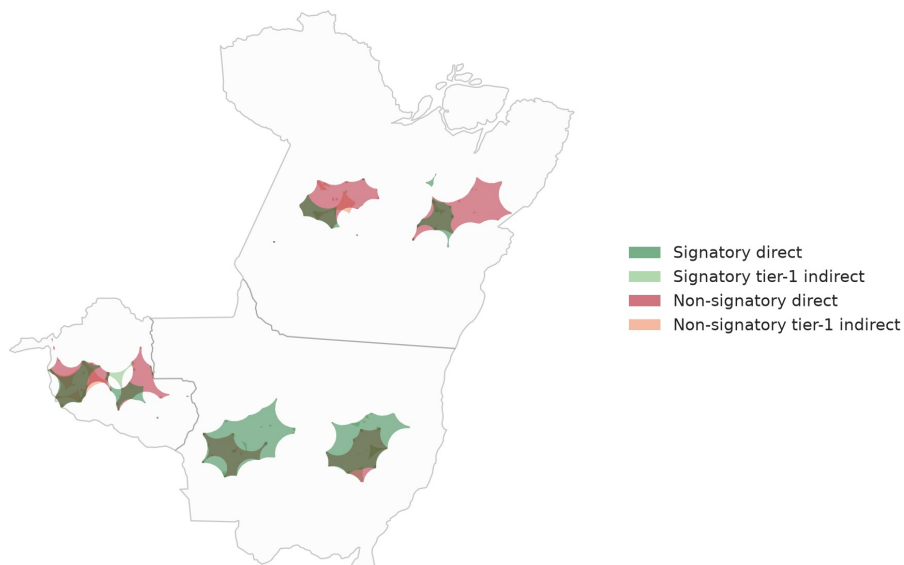


Figure 2. Cumulative overlap of all four supply-zone types, 2013-2018.

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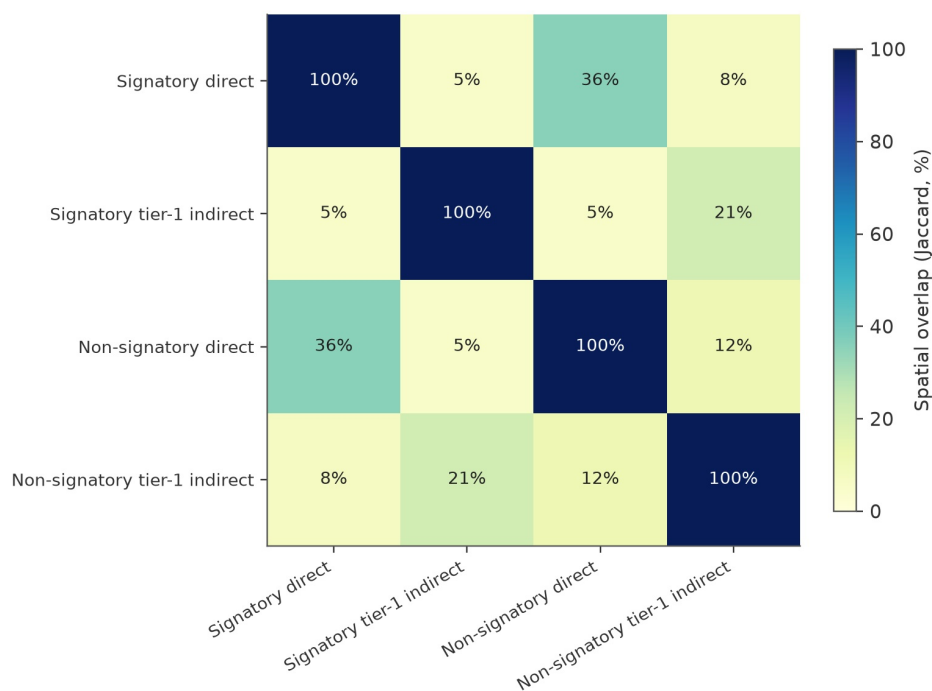


Figure 10. Pairwise spatial overlap between zone types, summarized as a Jaccard index.

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4. Does the same area stay in the supply zone every year

A property that supplies a plant only once is a weaker basis for monitoring than one that supplies it consistently. The workflow tracks, pixel by pixel, how many of the 6 years a location falls inside the signatory direct supply zone.

Years covered out of 6	Cumulative area (thousand ha)	Share of the ever-covered footprint (%)
1	750.0	4.8%
2	1,470.0	9.5%
3	1,460.0	9.4%
4	3,700.0	23.9%
5	4,320.0	27.9%
6	3,780.0	24.4%

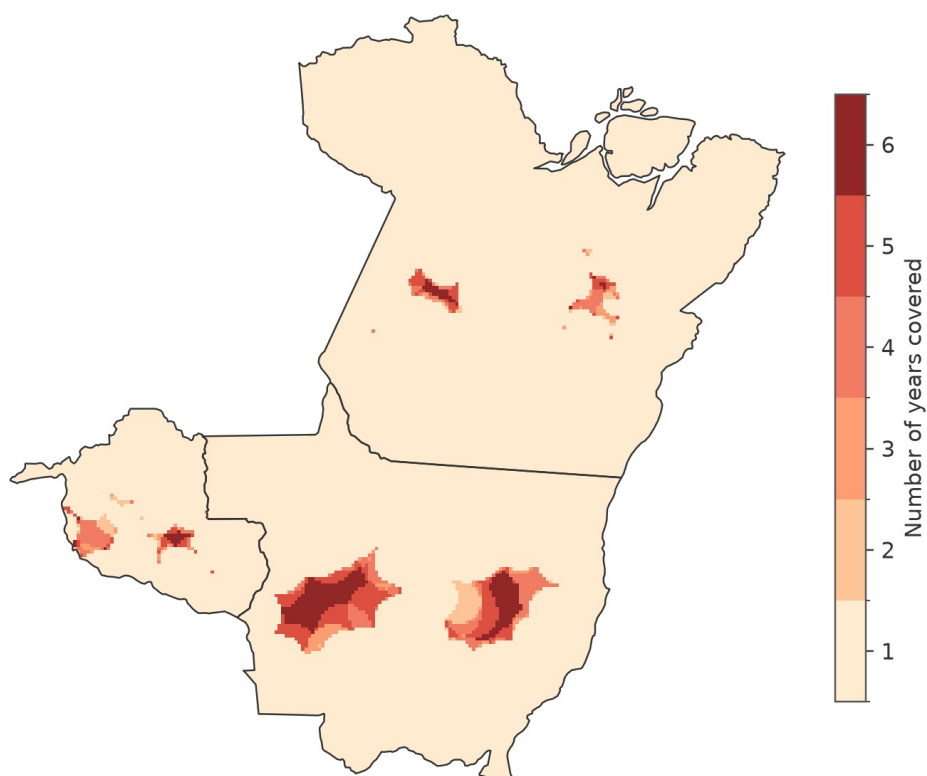


Figure 3. Number of years each location falls inside the signatory direct supply zone.

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5. What land cover is inside each supply zone

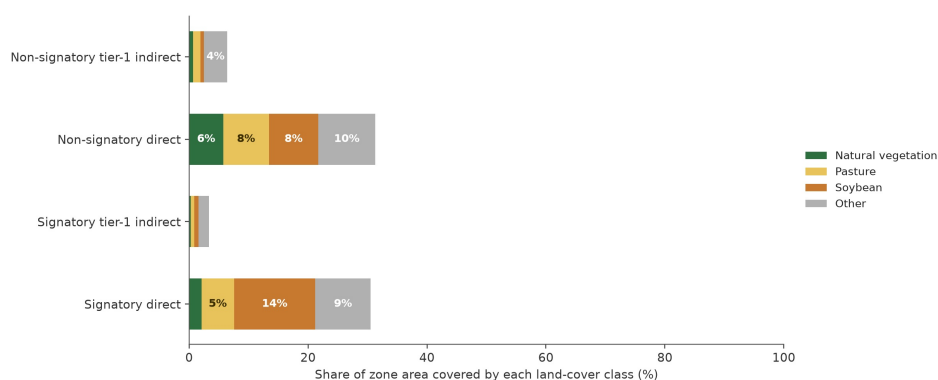


Figure 4. Land-cover composition inside each supply-zone type.

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Natural vegetation, pasture, soybean cropland, and a residual “other” class are cross-tabulated against each zone type, after excluding officially protected and military areas from the denominator for the natural-vegetation figure so that it reflects land that is legally available for conversion.

6. Deforestation and carbon inside the signatory direct zone

Zone type	Deforestation, 2008-2018 (thousand ha)	Committed emissions, 2008- 2018 (MtCO ₂ e)
Signatory direct	1.2	0.5
Signatory tier-1 indirect	0.5	0.0
Non-signatory direct	2.4	0.9
Non-signatory tier-1 indirect	0.5	0.0

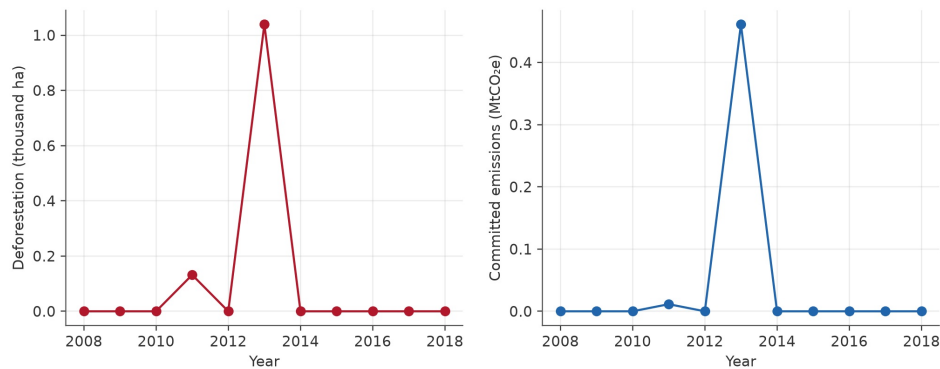


Figure S2. Annual synthetic deforestation and committed carbon emissions inside the signatory direct zone.

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7. How the supply-zone radius is chosen for each plant and year

Rather than applying one fixed radius to every plant, the workflow tests a range of distances for each plant-year and keeps the one at which cattle-volume-weighted spatial clustering is strongest. This makes zones larger around plants with a spatially concentrated supplier base and smaller around plants whose suppliers are scattered.

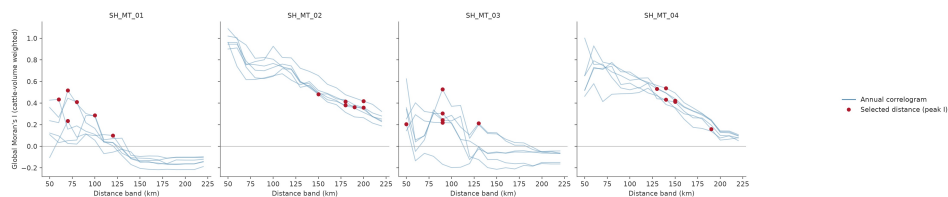


Figure 5. Correlogram of Moran's I against distance for four example slaughterhouses; the marked point is the distance actually used to build that plant-year's zone.

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8. Where does the cattle that leaves a signatory property end up

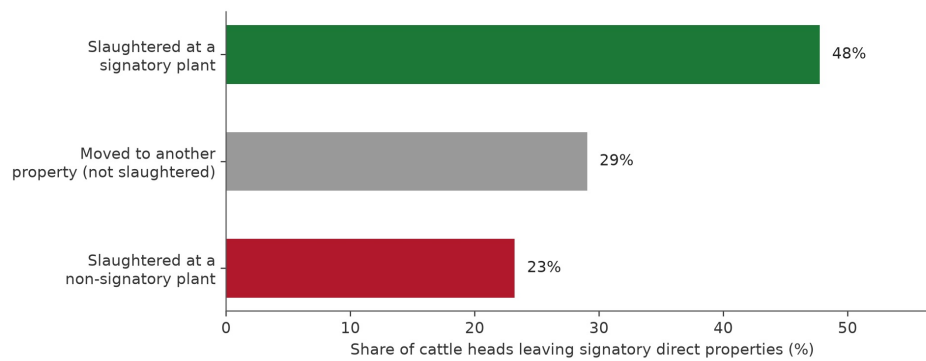


Figure 7. Destination of cattle heads that leave signatory direct properties.

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Only part of the cattle that leaves a signatory direct property is slaughtered at a signatory plant; the remainder is either slaughtered elsewhere or moved to another property first, which is exactly the kind of leakage that indirect-supplier monitoring is designed to catch.

9. How far apart are direct suppliers, indirect suppliers, and rival plants

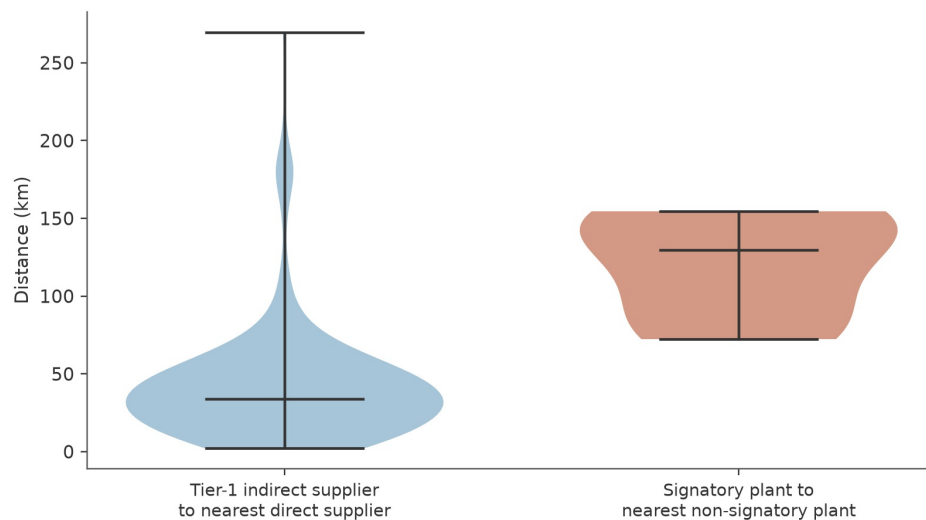


Figure 8. Distribution of two distance measures: tier-1 indirect supplier to nearest direct supplier, and signatory plant to nearest non-signatory plant.

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10. What would happen if monitoring were extended

Monitoring scenario	Zone area (million ha)	Properties covered	Slaughter volume covered (%)
Current (signatory direct only)	15.6	115	54.8%
Add signatory tier-1 indirect	16.1	127	60.3%

Add non-signatory direct	23.8	168	77.6%
All direct and tier-1 suppliers	24.2	168	77.6%

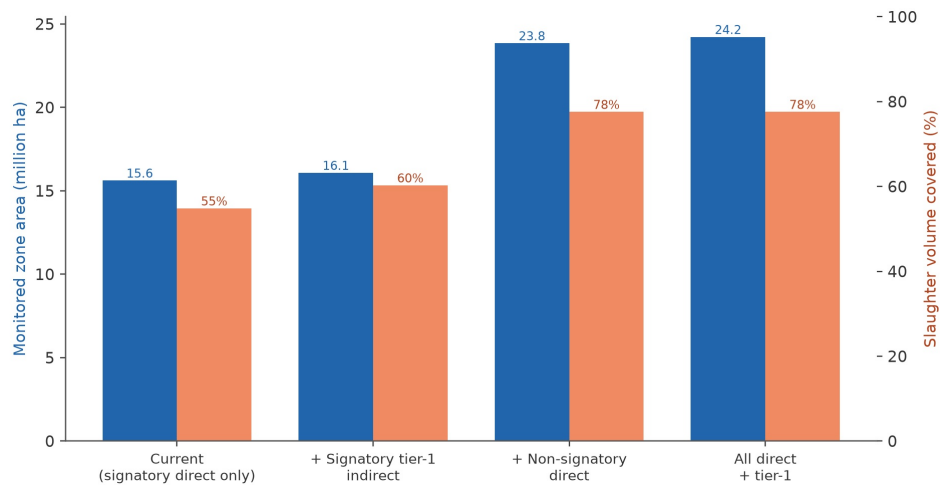


Figure 6. Monitored area and slaughter-volume coverage under four expansion scenarios.

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Adding non-signatory direct suppliers to the monitored footprint captures far more of the slaughter volume in this synthetic scenario than adding tier-1 indirect suppliers of signatory plants does, but it also requires monitoring a substantially larger area, illustrating the coverage-versus-scope trade-off that any traceability system has to navigate.

11. Does the choice of method matter

Method	Mean zone area (million ha)	Median overestimate vs. this workflow (%)
Incremental spatial-autocorrelation (this workflow)	1.7	-
Simple radial buffer around the plant	13.7	783.0%
Supplier convex-hull proxy	6.2	-

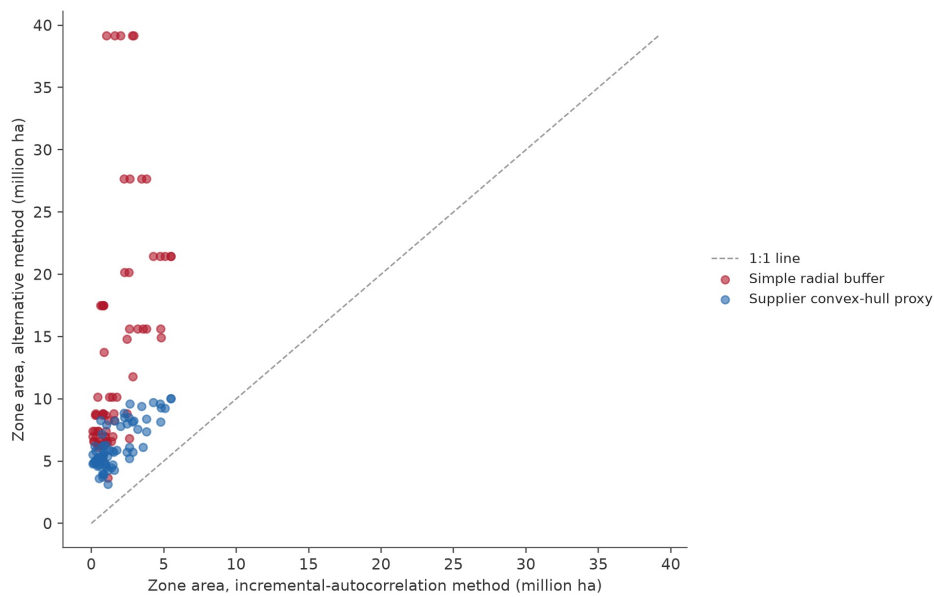


Figure 9. Zone area under the incremental spatial-autocorrelation method compared with two simpler distance-based proxies.

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Simple proxies, such as a fixed-radius buffer around the plant, tend to overestimate the true supply zone because they ignore the actual geographic spread of a plant's supplier base; the overestimate is largest, sometimes by more than an order of magnitude, in plant-years where the true supply zone is small and tightly clustered. The median is reported above rather than the mean because a handful of these small-zone plant-years produce extreme percentage overestimates that would otherwise dominate the average. This is a methodological caution for anyone tempted to shortcut the distance-selection step to save computation time.

12. Where the supply zone sits geographically, region by region

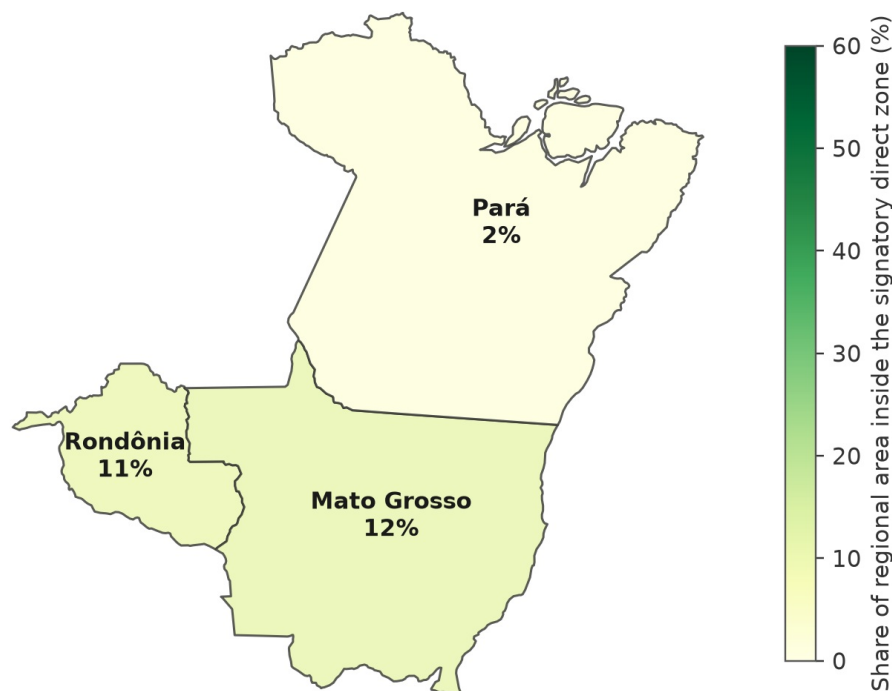


Figure 11. Share of each region's territory that falls inside the cumulative signatory direct supply zone.

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13. Limitations

- **All data are synthetic.** Every property boundary, transaction, and slaughterhouse in this repository was generated for testing purposes and carries no relationship to actual locations, companies, or individuals.
- **The open-source distance-selection and aggregation methods are declared analogues, not certified reproductions,** of the original study's ArcGIS Pro routines; see [docs/ARCGIS_EQUIVALENCE.md](#) for a line-by-line comparison of assumptions.
- **Land-use and deforestation layers are simplified** into four classes and a single annual raster; the original study's underlying data sources are richer.
- **Supplier-hull and radial-buffer comparisons are illustrative proxies** built for this repository; they should not be read as a critique of any specific commercial GIS workflow.

14. How to reproduce every number in this report

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -e ".[dev]"
python -m supply_zones all --clean
pytest
```

This regenerates the synthetic input data, reruns every analytical step described above,

rebuilds every figure and table referenced in this report (including this document itself), and runs the automated QA gates in `outputs/qa/QA_REPORT.md`.

Citation

Please cite the original article for the underlying method, and `CITATION.cff` in this repository for the synthetic reproduction code.